

Evaluating Usability, Cognitive Load, and Presence in VR Tutorials: A Preliminary Study

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Figure 1: Screenshots of exploratory and step-by-step onboarding VR activities.

ABSTRACT

The adoption of virtual reality (VR) in fields other than entertainment has resulted in non-VR users being exposed to immersive experiences. VR onboarding is essential to set up novice users for success. However, research pertaining VR onboarding is limited. This paper examines usability, cognitive load, and presence on exploratory (i.e., finding solutions through exploration) and step-by-step onboarding approaches found in some commercial video games and serious games developed for research. Our preliminary results indicate that exploratory onboarding is perceived more usable, cognitive load was similar between the two onboarding strategies, and presence was higher on exploratory onboarding.

Index Terms: Virtual reality, onboarding, user experience.

1 INTRODUCTION

Games and simulations can be effective training methods. Virtual reality (VR) is becoming a more common format for presenting these applications. VR expertise and proficiency are achieved through practice, which is driven by intrinsic motivation while maintaining engaging experiences that leverage flow [3]. Unlike video games, VR simulations, serious games, and other nonentertainment applications are not chosen by users but rather imposed as requirements for learning or certifications, among others. Moreover, the adoption of VR in non-entertainment scenarios has seen the technology being pushed to users lacking video game or VR expertise [5], such as nurses, caregivers, and tradeskill workers, to mention some. For such users, onboarding strategies are essential to set them up for success, while ensuring that relearning how to perform tasks in VR lacking proper real-world representation does not increase the cognitive load. This paper presents a preliminary

study on usability, cognitive load, and presence pertaining to VR onboarding across six different commercial and research-developed VR experiences. The rationale for selecting various commercial VR experiences was to identify common onboarding approaches.

2 MATERIALS AND METHODS

A within-subjects study approved by Ontario Tech University's Research Ethics Board #17597 required participants to complete the tutorials for six VR experiences consisting of four commercial video games (i.e., Beat Saber, Half Life Alyx, Kayak VR Mirage, and Job Simulator) and two serious games (i.e., Premium Cuts - developed in partnership with Meat and Poultry Ontario, and Snow Shovelling - developed for this study). The inclusion criteria were broad, only restricted to participants who were unable to use a VR headset. Conditions were randomized using Latin squares to minimize carryover effects. Due to VR headset and participant availability, a combination of Meta Quest 3 (MQ3) on PCVR for Snow Shovelling (SS) and Half Life Alyx (HLA), Meta Quest Pro (MQP) for Beat Saber (BS) and Premium Cuts (PC), and PlayStation VR (PSVR) for Job Simulator (JS) and Kayak VR Mirage (KVRM) were used for this study and explained in Sec. 2.2.

2.1 Study Procedures

Participants completed the following for each VR experience: i) a demographic survey that includes questions for VR self-efficacy (measured using the following questions in 5-point Likert scale format: I am comfortable navigating and interacting within virtual reality environments; I am confident in my ability to use VR controllers to manipulate objects; I have a good understanding of virtual reality hardware and software components; I am able to troubleshoot common issues that may arise while using virtual reality systems; and I am able to adapt to new virtual reality applications and environments quickly), ii) familiarizing themselves with the VR headset, iii) completing the tutorial for each VR experience, and iv) completing the System Usability Scale [1], the NASA Task Load Index [2], and the Presence Questionnaire (PQ) [4], followed by an open question to provide feedback.

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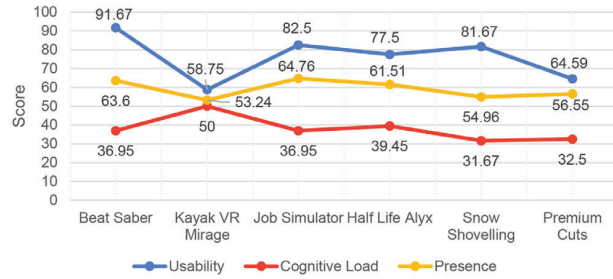


Figure 2: Usability, cognitive load, and presence results per VR experience.

2.2 Apparatus

BS is a rhythm game in which players slice cubes in the indicated direction and color, matching the saber. The step-by-step tutorial requires players to slice cubes and avoid obstacles until they have successfully completed. KVRM is a kayaking simulator with a step-by-step tutorial focused on how to paddle using both controllers while providing instructions using text and a ghost kayaker demonstrating the required movements. JS is a game with four jobs (i.e., gourmet chef, automotive mechanic, office worker, or store clerk) without a dedicated tutorial relying on exploration. The first three orders of the gourmet chef job were selected to act as the onboarding phase (i.e., cooking bacon and eggs, making a smoothie using a blender, and making tea and crumpets using a toaster and a kettle). HLA is an action-adventure/horror game with exploratory onboarding occurring throughout the beginning of the game; we chose the section explaining how to use gravity gloves to pick up objects from a distance. Tasks and encouragement are given orally by a character, in addition to a floating text following the player's hand. The two serious games include SS, which is a custom-made experience to remove snow from the ground and deposit it in a designated area, with an exploratory onboarding approach allowing players to choose how to use the shovel, and PC, which allows cutting meat with a step-by-step tutorial with spoken and text instructions that teach players how to clean the knife, wash their hands, sharpen a knife and cut the meat.

3 PRELIMINARY RESULTS

Six participants (male = 5, female = 1), one with a background in computer science and the remainder in game development, volunteered to participate in the study. Four participants considered themselves novice VR users, one intermediate and one advanced, who indicated playing BS on a weekly basis. Interestingly, despite the levels of expertise reported, the comfort and self-efficacy results indicated that four participants ranked moderate while two ranked as high (1 to 1.9: "Very Low," 2 to 2.9: "Low," 3 to 3.9: "Moderate," and 4 to 5: "High"). With respect to usability, the System Usability Scale (SUS) BS had the highest score ($M = 91.67$, $SD = 8.76$), followed by JS ($M = 82.50$, $SD = 17.90$), SS ($M = 81.67$, $SD = 13.00$), HLA ($M = 77.50$, $SD = 24.00$), PC ($M = 64.59$, $SD = 19.8$), and KVRM ($M = 58.75$, $SD = 34.20$). In terms of cognitive load, the raw NASA TLX indicated that SS had the lowest score ($M = 21.40$, $SD = 6.74$), followed by PC ($M = 32.50$, $SD = 14.18$), BS ($M = 36.95$, $SD = 8.39$), JS ($M = 36.95$, $SD = 8.05$), HLA ($M = 39.45$, $SD = 14.24$), and KVRM ($M = 50.00$, $SD = 16.63$). Finally, in terms of presence, the results of the presence questionnaire indicated that JS had the highest score ($M = 64.76$, $SD = 2.85$), followed by BS ($M = 63.60$, $SD = 5.23$), HLA ($M = 61.51$, $SD = 7.65$), PC ($M = 56.55$, $SD = 4.96$), SS ($M = 54.96$, $SD = 12.09$) and KVRM ($M = 53.24$, $SD = 12.10$). The results are presented in Fig. 2.

4 DISCUSSION

This paper presented a preliminary user study that focused on evaluating usability, cognitive load, and presence associated with onboarding approaches in six VR experiences. Onboarding, regardless of experience, and VR technology should be highly usable, with little cognitive load and high presence, to set users up for success. Although our current results are limited due to the small number of participants and their backgrounds, we believe that this first step has provided insights and future research directions that can inform practical applications for developers and educators. Regarding usability, scores above 68 are considered acceptable, above 80 good, and above 90 excellent. Although two exploratory onboarding approaches (i.e., JS and SS) had high usability, it was BS with a step-by-step tutorial that had the highest SUS. In contrast, difficulties in completing the tasks with KVRM when paddling and relocating after respawning outside of the training area made this game the least usable. Additionally, PC's usability was negatively impacted by inconsistencies experienced when following some of the instructions and a reported bug that required restarting the tutorial caused in the hand washing stage. More research is needed to better understand what specific interactions, game mechanics, and task representation influenced this result.

In terms of cognitive load, scores between 0 and 29 indicate low to moderate, 30-49 somewhat high, and above 50 high load. Interestingly, BS did not have the lowest cognitive load, which we believe was due to tasks requiring participants to focus on evading obstacles and slicing direction while color matching the saber and cubes. However, SS had the lowest usability given the intuitive interaction with the shovel, followed by PC with the second lowest cognitive load, which we believe is due to the appropriate representation of tasks and guided instructions. With regard to presence, JS offered the highest possibility to act, realism of the tasks, and the self-evaluation of performance, while KVRM was the worst rated game in all accounts, having the lowest usability, the highest cognitive load, and the lowest presence caused by frustration and physical demand that affected actions. It is worth noting that KVRM and HLA had the highest visual realism, but it was JS with a toon shader style that had the highest realism. Therefore, our study highlights that graphical realism is not the only factor for higher presence and how the actions are represented matters.

Future work will conduct a larger user study with diverse participants, in addition to comparing applications that showcase similar onboarding approaches and VR headsets. Performance metrics and gameplay recordings will be collected to identify elements that influence the user experience. Such future work will inform novel forms of successful onboarding.

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